THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VII — SCIENCE · ANNUAL / FULL-YEAR PRACTICE PAPER

Session 2026-27 · Set 01 · Emphasis: Materials & Chemical Changes

Class: 7	Subject: Science	Set 01	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

SECTION A — Multiple Choice Questions — 1 questions · 12 marks

Question 1 has twelve parts (I–XII). Each carries 1 mark.

1. From Question I–XII, choose the correct option.

I. A paper strip stained with turmeric is dipped in a soap solution. The strip turns: 1

(a) green (b) red (c) colourless (d) blue

II. The heating element of an electric heater is made of nichrome because nichrome has: 1

(a) high resistance and low melting point (b) low resistance and low melting point

(c) high resistance and high melting point (d) zero resistance

III. Which of the following non-metals is a good conductor of electricity? 1

(a) Iodine (b) Phosphorus (c) Sulphur (d) Graphite

IV. Which of the following is a chemical change? 1

(a) Dissolving sugar in water (b) Tearing of paper

(c) Melting of wax (d) Curdling of milk

V. A person's body does not make enough insulin. The gland that is affected is the: 1

(a) pituitary (b) thyroid (c) adrenal (d) pancreas

VI. Water in a pan on a stove is heated from below. The heat spreads through the water mainly by: 1

(a) none of these (b) radiation only (c) conduction (d) convection

VII. A car covers 150 km in 2 h 30 min. Its average speed is: 1

(a) 75 km/h (b) 50 km/h (c) 60 km/h (d) 65 km/h

VIII. Most of the digested food is absorbed into the blood in the: 1

(a) large intestine (b) small intestine

(c) oesophagus (d) stomach

IX. Plants take in carbon dioxide for photosynthesis through tiny pores in the leaf called: 1

(a) stomata (b) veins (c) xylem vessels (d) root hairs

X. The image formed by a plane mirror is: 1

(a) real, inverted and of the same size

(b) virtual, upright, of the same size and laterally inverted

(c) virtual and diminished

(d) real and magnified

XI A solar eclipse can occur only on: 1

(a) a half-moon day (b) any day the Moon is bright

(c) a new moon day (d) a full moon day

XII Zinc granules are added to dilute hydrochloric acid and a burning matchstick is brought near the mouth of the test tube. A 'pop' sound is heard because the gas released is: 1

(a) oxygen (b) carbon dioxide (c) hydrogen (d) nitrogen

SECTION B — Very Short Answer Questions — 4 questions · 8 marks

2 marks each (about 30–40 words).

2. Farmers spread slaked lime on soil which has become too acidic. Name the type of reaction involved and state its result in one line. **2**

OR

Name two natural indicators. How would turmeric and blue litmus behave when dipped in lemon juice?

3. Electric wires are made of copper but are covered with plastic. Give the reason for each choice. **2**

4. Classify each as a physical or chemical change: (a) melting of ice (b) rusting of an iron gate (c) dissolving salt in water (d) burning of a candle. **2**

5. The Moon always shows the same face to us. Give the reason. **2**

SECTION C — Short Answer Questions — 3 questions · 9 marks

3 marks each (about 60 words).

6. (a) Write the word equation for the burning of magnesium ribbon in air. (b) A clean iron nail is kept in copper sulphate solution for 30 minutes. State two observations. **3**

OR

Rusting is a serious problem for iron bridges and railings. (a) Name the two things needed for rusting. (b) Give two methods to prevent it.

7. Describe a simple activity to show that a gas is produced when baking soda is mixed with vinegar. Name the gas and state how you will test it. **3**

8. A bulb in a house stops working suddenly and the fuse has blown. (a) What is a short circuit? (b) How does a fuse protect appliances? (c) Name a modern device that replaces the fuse. **3**

SECTION D — Long Answer Questions — 3 questions · 15 marks

5 marks each.

9. Rina tested four colourless solutions P, Q, R and S with indicators. Her observations are in the table. **5**

Solution	Observation
P	Blue litmus turns red
Q	Red litmus turns blue; turmeric turns red
R	No change in red or blue litmus or turmeric
S	Red litmus turns blue

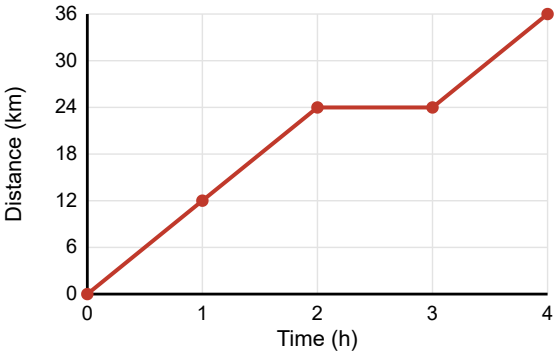
(i) Classify P, Q, R and S as acidic, basic or neutral. (ii) Name one household substance which could be R. (iii) If P and S are mixed in the right amount, what type of reaction takes place and what is formed?

10. (a) Differentiate between metals and non-metals on any four properties (lustre, malleability/ductility, sound and conduction). (4) **5**
(b) Name one metal that is a liquid at room temperature. (1)

OR

(a) Describe an experiment to find out the conditions needed for rusting using three test tubes. (3)
(b) In which test tube will rust form fastest and why? (2)

11. The graph shows the journey of a cyclist. Use it to answer: (i) Find the speed during the first 2 hours. (ii) What does the horizontal part between 2 h and 3 h show? (iii) What distance was covered in 4 h? (iv) Find the average speed for the whole journey. (v) Is the motion uniform or non-uniform? **5**



OR

(a) Explain conduction, convection and radiation with one everyday example each. (3)
(b) Why do we prefer to wear light-coloured clothes in summer? (1)
(c) Name the instrument used to measure human body temperature and write its range. (1)

SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

Read the cases carefully and answer. Each question carries 4 marks.

12. Meera, who is 13, tells her friend that her height has increased quickly in the past year, she has begun to feel moody, and she is worried about her skin developing pimples. Her friend Arjun says his voice has become deep and cracks sometimes.

(a) Name the stage of life in which these changes occur. 1
(b) Which structure in the throat grows larger in boys and causes the voice to deepen? 1
(c) Name two chemical messengers that bring these changes and the glands that release them. 2

OR

(d) Suggest two healthy habits for adolescents.

13. Ravi had a heavy lunch of dal, roti, rice and curd. After some time he felt his heart beating fast while playing, and his breathing became deeper.

(a) In which organ does the digestion of proteins begin? 1
(b) Which blood cells carry oxygen, and which pigment helps in this? 1
(c) Explain why breathing and heartbeat rates increase during exercise. 2

OR

(d) Write the word equation of cellular respiration.

14. In a school experiment, Shreya covered a part of a leaf of a potted geranium plant with black paper and kept it in sunlight for a day. She then boiled the leaf in alcohol and tested it with iodine.

(a) What is the purpose of boiling the leaf in alcohol? 1
(b) Which part of the leaf will turn blue-black with iodine? 1
(c) What does the result show about the conditions for photosynthesis? 2

OR

(d) Name the gas given out and the raw material taken in by leaves during photosynthesis.

15. A bus conductor sits near the rear-view mirror and a dentist uses a small curved mirror to see teeth. A student makes a pinhole camera from a shoe box and observes a candle flame through it.

(a) What type of mirror is used as a rear-view mirror in vehicles? 1

(b) State the nature of the image seen in the pinhole camera. 1

(c) Why is a convex mirror preferred as a rear-view mirror? 2

OR

(d) Why does the dentist use a concave mirror?